

Tushar Patwal

DATA SCIENTIST

Delhi, India • +91 7678331903 • patwaltushar001@gmail.com

SUMMARY

Data Scientist with 2+ years of experience in Python, SQL, machine learning, data preprocessing, exploratory data analysis, feature engineering, statistical analysis, and AI/ML application development. Experienced in building production-grade data pipelines, REST APIs, LLM applications, and scalable AI workflows using Python, Pandas, NumPy, Scikit-learn, FastAPI, SQL Server, Docker, LangGraph, OpenAI, and Gemini.

PROFESSIONAL EXPERIENCE

Data Scientist, ChemAnalyst Data Pvt Ltd.

Oct 2025 - Present

1. Developed and deployed a production-grade REST API using Python, FastAPI, Pydantic, and SQL Server for natural-language query processing, integrating AI capabilities with graceful fallback mechanisms.
2. Designed an asynchronous batch processing platform using Python, Streamlit, Redis, RQ workers, Docker, and Docker Compose for scalable job execution, real-time progress tracking, and fault tolerance.
3. Orchestrated a multi-service application stack comprising Streamlit, Redis, RQ workers, and SQL Server using Docker Compose, with structured logging and JSON-based batch progress tracking.
4. Developed an ensemble machine learning model combining Random Forest, XGBoost, Decision Tree, and CatBoost to forecast chemical product prices for upcoming months. Applied time-series cross-validation and SHAP analysis to evaluate model performance and identify key factors influencing price predictions, achieving 73% prediction accuracy and supporting data-driven procurement planning.

Automation Engineer, Info Edge India Ltd

July 2024 - July 2025

1. Developed and automated ETL pipelines processing 600K+ records monthly using Python and SQL, implementing data validation, error handling, and quality checks to improve data reliability.
2. Built LLM inference pipelines with retry logic, rate limiting, and error tracking to maintain reliability under high-volume workloads.
3. Executed weekly batch updates for 700+ partners, resolving data discrepancies and enforcing strict schema adherence for user-facing content.

TECHNICAL SKILLS

Programming: Python, SQL, HTML, CSS, JSON

Data Science & ML: Pandas, NumPy, Scikit-learn, Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Machine Learning

Generative AI & LLM: Generative AI, LLMs, OpenAI API, Google Gemini API, LangChain, LangGraph, RAG, Prompt Engineering, Embeddings, AI Agents

Backend & APIs: FastAPI, REST APIs, Pydantic, API Development, Request/Response Modeling

Databases & Vector Search: SQL Server, PostgreSQL, MySQL, FAISS, PyODBC

Data Engineering: ETL Pipelines, Data Processing, Data Validation, Batch Processing, Asynchronous Processing

DevOps & Tools: Docker, Docker Compose, Redis, RQ, Git, GitHub, Linux, VS Code

EDUCATION

Bachelor of Computer Applications (BCA) – Artificial Intelligence & Data Science

Vivekananda Institute of Professional Studies | CGPA: 9

Senior Secondary Education – CBSE | 76.4%

Secondary Education – CBSE | 75.9%

PROJECTS

1. Chemical Market Chatbot | Python, FastAPI, SQL Server, spaCy, OpenAI API, Pydantic

- Developed a production-ready REST API with a POST /api/chat endpoint for natural-language querying of SQL Server data. Implemented NLP and semantic processing using spaCy and sentence-transformers, with OpenAI integration and graceful fallback for reliable responses.

2. Landing Page Generator | Python, Streamlit, Redis, RQ, Docker, SQL Server

- Built a containerized asynchronous batch platform using Redis as a broker and RQ workers for scalable job execution and real-time progress tracking.
- Orchestrated application, Redis, and worker services with Docker Compose, adding structured logging and JSONbased batch monitoring for debugging.

3. External News Aggregator | Python, Docker, Docker Compose, SQL Database, SearXNG

- Developed a Dockerized data collection service with database schema validation and modular search, reporting and storage components.